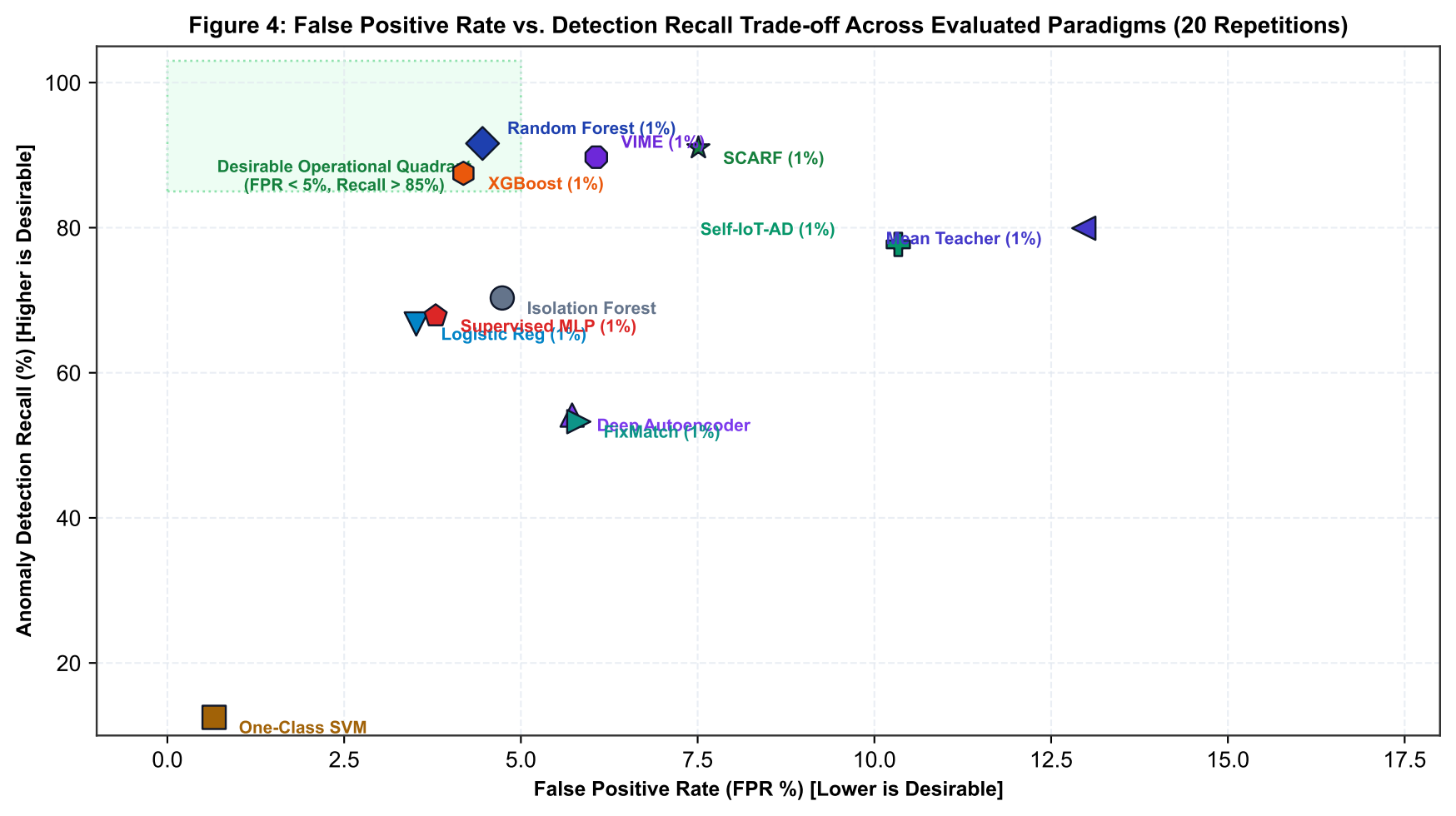
Figure 4: False Positive Rate vs. Detection Recall Trade-off Across Evaluated Paradigms (20 Repetitions)
Anomaly Detection Recall (%) [Higher is Desirable]
False Positive Rate (FPR %) [Lower is Desirable]
100
80
60
40
20
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Desirable Operational Quadrant
(FPR < 5%, Recall > 85%)
Random Forest (1%)
VIME (1%)
SCARF (1%)
XGBoost (1%)
Self-IoT-AD (1%)
Mean Teacher (1%)
Isolation Forest
Supervised MLP (1%)
Logistic Reg (1%)
Deep Autoencoder
FixMatch (1%)
One-Class SVM